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arta med dagskarta

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Image file:  65c Borrhålskarta med dagskarta.jpg
Image size:  10905 × 7104 pixels
Number of GCPs:  16

Source CRS:  EPSG:3152  (ST74, Stockholm 1938)
Target CRS:  EPSG:3006  (SWEREF99 TM)

Transformation:  pyproj (PROJ pipeline)
Pipeline:  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808

Inter-consistency (affine fit):
  RMS residual:  1.062 m
  Max residual:  3.312 m
  Outlier threshold (2×RMS):  2.124 m

Overall quality:  GOOD
  (1 outlier GCP(s) detected: 5)
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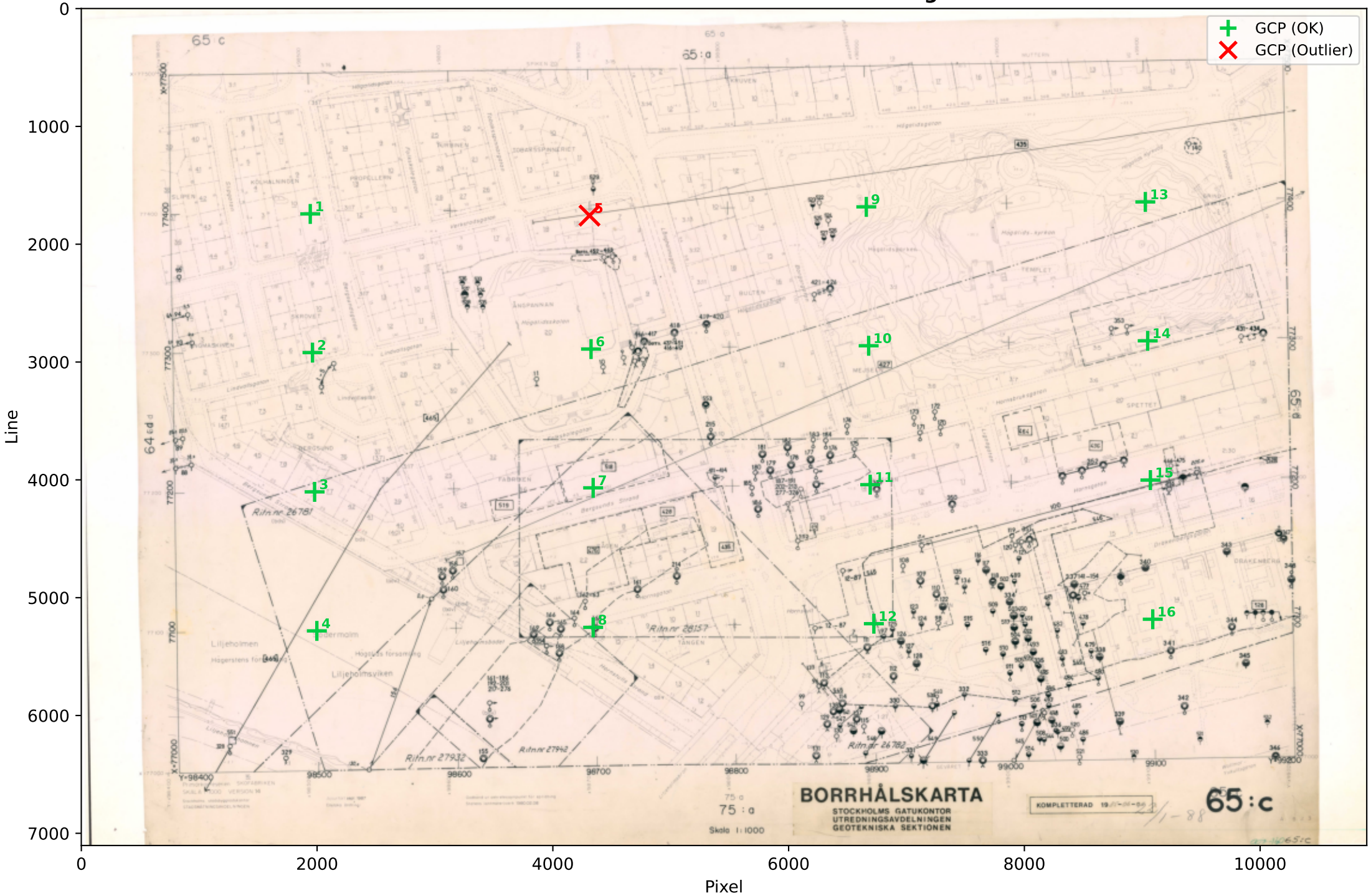
GCP Locations — 65c Borrhålskarta med dagskarta



Inter-consistency Analysis — Affine Fit Residuals

GCP	Pixel	Line	ST74 X	ST74 Y	Res. X (m)	Res. Y (m)	Res. Total (m)	Flag
1	1942	1743	98500	77400	-0.062	-0.833	0.835	
2	1961	2920	98500	77300	-0.158	-0.802	0.818	
3	1979	4102	98500	77200	-0.163	-0.348	0.384	
4	1998	5282	98500	77100	-0.255	-0.063	0.263	
5	4311	1757	98700	77400	-0.583	3.260	3.312	OUTLIER
6	4324	2890	98700	77300	-0.228	-0.455	0.509	
7	4342	4068	98700	77200	-0.238	-0.341	0.416	
8	4342	5252	98700	77100	1.283	0.261	1.309	
9	6659	1683	98900	77400	0.560	-0.150	0.580	
10	6679	2862	98900	77300	0.382	0.052	0.385	
11	6692	4041	98900	77200	0.796	0.245	0.833	
12	6723	5221	98900	77100	-0.312	0.545	0.628	
13	9026	1641	99100	77400	0.136	-0.817	0.829	
14	9046	2820	99100	77300	-0.042	-0.616	0.617	
15	9069	4002	99100	77200	-0.470	-0.155	0.495	
16	9089	5183	99100	77100	-0.646	0.216	0.681	

```
Affine model:  map_x = 0.084651·pixel + -0.001285·line + 98337.91
               map_y = -0.001226·pixel + -0.084968·line + 77551.31

RMS residual: 1.062 m    |    Max residual: 3.312 m    |    Outlier threshold (2×RMS): 2.124 m

Grid spacing (map coords):  ΔX = [200.0, 200.0, 200.0]    ΔY = [100.0, 100.0, 100.0]
Pixel spacing (adjacent columns):  mean=2362.5  std=10.9  range=[2344, 2381]
Line spacing (adjacent rows):  mean=1176.2  std=13.1  range=[1133, 1184]
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Coordinate Transformation — EPSG:3152 → EPSG:3006

GCP	ST74 X	ST74 Y	SWEREF99TM E	SWEREF99TM N
1	98500	77400	672340.456	6579339.278
2	98500	77300	672345.045	6579239.387
3	98500	77200	672349.634	6579139.495
4	98500	77100	672354.223	6579039.603
5	98700	77400	672540.239	6579348.456
6	98700	77300	672544.828	6579248.565
7	98700	77200	672549.417	6579148.673
8	98700	77100	672554.006	6579048.781
9	98900	77400	672740.023	6579357.635
10	98900	77300	672744.612	6579257.743
11	98900	77200	672749.201	6579157.851
12	98900	77100	672753.790	6579057.959
13	99100	77400	672939.806	6579366.813
14	99100	77300	672944.395	6579266.921
15	99100	77200	672948.984	6579167.029
16	99100	77100	672953.573	6579067.137

```
Source CRS:  EPSG:3152  —  ST74, Stockholm 1938 (local TM)
Target CRS:  EPSG:3006  —  SWEREF99 TM (ETRS89 / UTM zone 33N)

PROJ pipeline:
  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808 y_0=-6500614.7836 ellps=GRS80 step proj=utm zone=33 ellps=GR

This is equivalent to Lantmäteriet GTRANS.  The pipeline performs:
  1. Inverse ST74 Transverse Mercator projection (→ geographic on GRS80)
  2. Forward UTM zone 33 projection (→ SWEREF99TM)

No datum shift is required — both CRS use GRS80 ellipsoid and
SWEREF99/ETRS89 realization.  Round-trip accuracy: < 0.001 mm.
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